

Downloading the Sawtooth configuration file

Download the Kubernetes configuration file for a single-node environment from https://sawtooth.hyperledger.org/docs/core/nightly/master/app_developers_guide/sawtooth-kubernetes-default.yaml.

The `sawtooth-kubernetes-default.yaml` file defines the process for constructing a one-node Sawtooth environment with the following containers:

- A single validator using dev mode consensus
- A REST API connected to the validator
- The *Settings* transaction processor (`sawtooth-settings`)
- The *IntegerKey* transaction processor (`intkey-tp-python`)
- The *XO* transaction processor (`xo-tp-python`)
- A shell container for running Sawtooth commands (a command line client)

The configuration file also specifies the container images to download (from DockerHub) and the network settings needed for the containers to communicate correctly.

Starting the Sawtooth cluster

Use the following steps to start the Sawtooth network, and change your working directory to the same one in which you saved the configuration file. Make sure that Minikube is running.

```
$ minikube status
minikube: Running
cluster: Running
kubectl: Correctly Configured: pointing to minikube-vm at 192.168.99.100
```

Minikube start

Start Sawtooth in a local Kubernetes cluster, as follows:

```
$ kubectl apply -f sawtooth-kubernetes-default.yaml
```

Optionally, start the Minikube dashboard using the following command:

```
$ minikube dashboard
```

This command opens the dashboard in your default browser. The overview page shows the Sawtooth deployment (`sawtooth-0`) and pod (`sawtooth-0-POD-ID`).

To test basic Sawtooth functionality, connect to the shell container using the command below:

```
$ kubectl exec -it $(kubectl get pods | awk '/sawtooth-0/{print $1}') --container sawtooth-shell -- bash
```

To display the list of blocks on the Sawtooth blockchain, use the command given below:

```
root@sawtooth-0# sawtooth block list
```

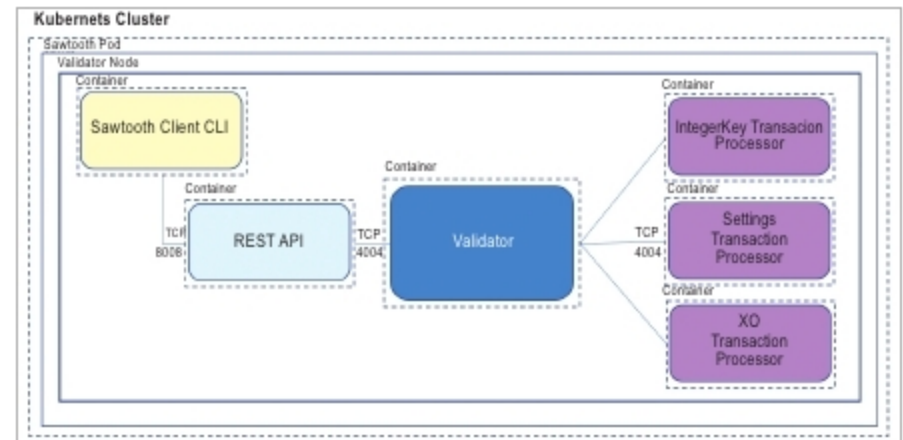


Figure 1: Sawtooth Kubernetes deployment architecture

Because this is a new blockchain, there is only one block. Block 0 is the genesis block. The output is similar to the example shown below:

```
NUM  BLOCK_ID  BATS  TXNS  SIGNER
020d7b66577217 58d1ad1a3392daadd57473 d84e1e1c8c58c14ec8
62ff7fbf44a3bef4d82c 40052dd8fc2808191f83
0447df59fe074aea02a000ff64bc458e256 1 1 025f80...
```

Copy the block ID from the previous output, and then use the following command to display more information about the block.

```
root@sawtooth-0# sawtooth block show {BLOCK-ID}
```

The output of this command is quite long, because it includes all data stored under that block.

```
batches:
- header:
  signer_public_key: 03f257dee6f021b579 cb59d34f2489603892
  d44bb2e181eaa444e1bb4f4b4b812d
  transaction_ids:
  -
  3f6c2f60a66317 f09d052757dba605d 0c1c56caa38cdfdefb d7f45
  11a830a1fc22d8e 13ff86201ac309344 605b5df77a85e5979 9c16c
  3ba9e3cba950b709be04
  header_signature:
  6e5446e99bae1fe2d7d4a7561880bd069cc 404e099dd4380a
  7f491dd0588584b0b6b558d636eb42720d6c 839c6755182d30
  04b905429088413df00f82ec0fd1e
  ...
```

At this point, your environment is ready for experimenting with Sawtooth. The rest of this section introduces you to Sawtooth functionality.

Verifying the Sawtooth components

To check whether a Sawtooth component is running, connect to the component's container and run the `ps` command.

Use the `kubectl exec` command from your computer to connect to a Sawtooth container. On the Kubernetes dashboard,